

CS 2601 Linear and Convex Optimization

2. Math review

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Outline

- First-order conditions for unconstrained local min
- Second-order conditions for unconstrained local min

Review: Derivative

$\mathbf{x}$ is an **interior point** of $X \subset \mathbb{R}^n$ if there exists $\epsilon > 0$ s.t. $B(\mathbf{x}, \epsilon) \subset X$.

The **interior** of X , denoted by $\text{int } X$, is the set of interior points of X .

A function $f : X \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **differentiable** at $\mathbf{x}_0 \in \text{int } X$, if there exists a matrix¹ $A \in \mathbb{R}^{m \times n}$ s.t.

$$\lim_{\Delta \mathbf{x} \rightarrow \mathbf{0}} \frac{f(\mathbf{x}_0 + \Delta \mathbf{x}) - f(\mathbf{x}_0) - A\Delta \mathbf{x}}{\|\Delta \mathbf{x}\|} = \mathbf{0}$$

i.e.

$$\Delta f := f(\mathbf{x}_0 + \Delta \mathbf{x}) - f(\mathbf{x}_0) = A\Delta \mathbf{x} + o(\|\Delta \mathbf{x}\|)$$

The affine function $f(\mathbf{x}_0) + A(\mathbf{x} - \mathbf{x}_0)$ is the first-order approximation of f at $\mathbf{x}_0$,

$$f(\mathbf{x}) = f(\mathbf{x}_0) + A(\mathbf{x} - \mathbf{x}_0) + o(\|\mathbf{x} - \mathbf{x}_0\|)$$

¹More precisely, a linear transformation represented by matrix A

Review: Derivative

The matrix A is called the **derivative** of f at $\mathbf{x}_0$, and we write

$$\mathbf{f}'(\mathbf{x}_0) = D\mathbf{f}(\mathbf{x}_0) = A$$

The derivative is given by the **Jacobian matrix** of $\mathbf{f} = (f_1, \dots, f_m)$

$$\mathbf{f}'(\mathbf{x}_0) = \begin{bmatrix} \frac{\partial f_1(\mathbf{x}_0)}{\partial x_1} & \frac{\partial f_1(\mathbf{x}_0)}{\partial x_2} & \cdots & \frac{\partial f_1(\mathbf{x}_0)}{\partial x_n} \\ \frac{\partial f_2(\mathbf{x}_0)}{\partial x_1} & \frac{\partial f_2(\mathbf{x}_0)}{\partial x_2} & \cdots & \frac{\partial f_2(\mathbf{x}_0)}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m(\mathbf{x}_0)}{\partial x_1} & \frac{\partial f_m(\mathbf{x}_0)}{\partial x_2} & \cdots & \frac{\partial f_m(\mathbf{x}_0)}{\partial x_n} \end{bmatrix}$$

i.e.

$$[\mathbf{f}'(\mathbf{x}_0)]_{ij} = \frac{\partial f_i(\mathbf{x}_0)}{\partial x_j}, \quad i = 1, \dots, m; j = 1, \dots, n$$

Note

$$f_i(\mathbf{x}_0 + \Delta \mathbf{x}) = f_i(\mathbf{x}_0) + \sum_{j=1}^n \frac{\partial f_i(\mathbf{x}_0)}{\partial x_j} \Delta x_j + o(\|\Delta \mathbf{x}\|), \quad i = 1, 2, \dots, m$$

Review: Derivative

Example. An affine function $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ has derivative $f'(\mathbf{x}) = \mathbf{A}$ at all $\mathbf{x}$. In particular, when $m = 1$, $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} + b$ has derivative $f'(\mathbf{x}) = \mathbf{a}^T$, which is a $1 \times n$ matrix, i.e. a row vector.

Proof. In component form,

$$f_i(\mathbf{x}) = \sum_{k=1}^n A_{ik}x_k + b_i = A_{i1}x_1 + A_{i2}x_2 + \cdots + A_{in}x_n + b_i$$

so

$$\frac{\partial f_i(\mathbf{x}_0)}{\partial x_j} = A_{ij} \implies f'(\mathbf{x}_0) = \mathbf{A}$$

Alternative proof.

$$f(\mathbf{x}_0 + \Delta \mathbf{x}) - f(\mathbf{x}_0) = \mathbf{A} \Delta \mathbf{x} \implies f'(\mathbf{x}_0) = \mathbf{A}$$

Review: Derivative

Example. For symmetric A , $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$ has derivative

$$f'(\mathbf{x}) = 2\mathbf{x}^T A$$

Proof.

$$\frac{\partial f}{\partial x_k} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} \left(x_j \frac{\partial x_i}{\partial x_k} + x_i \frac{\partial x_j}{\partial x_k} \right) = \sum_{j=1}^n A_{kj} x_j + \sum_{i=1}^n A_{ik} x_i = 2 \sum_{i=1}^n x_i A_{ik}$$

Alternatively,

$$f(\mathbf{x}_0 + \Delta \mathbf{x}) - f(\mathbf{x}_0) = \mathbf{x}_0^T (A + A^T) \Delta \mathbf{x} + \underbrace{\Delta \mathbf{x}^T A \Delta \mathbf{x}}_{=o(\|\Delta \mathbf{x}\|)}$$

Note. For general A , $f'(\mathbf{x}) = \mathbf{x}^T (A + A^T)$. This can also be obtained by noting $\mathbf{x}^T A \mathbf{x} = \mathbf{x}^T \tilde{A} \mathbf{x}$ and $f'(\mathbf{x}) = 2\mathbf{x}^T \tilde{A}$, where $\tilde{A} = \frac{1}{2}(A + A^T)$.

Review: Gradient

For a real-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **gradient** of f at $\mathbf{x}$, denoted by $\nabla f(\mathbf{x})$, is the transpose of $f'(\mathbf{x})$,

$$\nabla f(\mathbf{x}) = [f'(\mathbf{x})]^T, \quad [\nabla f(\mathbf{x})]_i = \frac{\partial f(\mathbf{x})}{\partial x_i}, \quad i = 1, \dots, n$$

$\nabla f(\mathbf{x})$ is a column vector and satisfies

$$f'(\mathbf{x})\Delta\mathbf{x} = \langle \nabla f(\mathbf{x}), \Delta\mathbf{x} \rangle = \nabla f(\mathbf{x})^T \Delta\mathbf{x}$$

The first-order approximation of f at $\mathbf{x}_0$ is

$$f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0)$$

Example. For symmetric $\mathbf{A}$, the gradient of $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ is

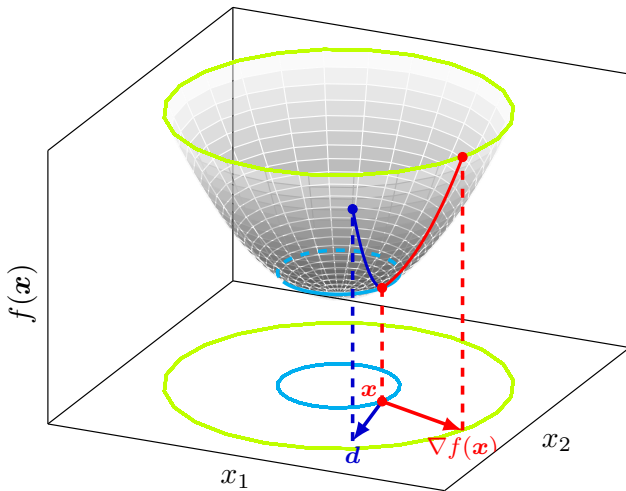
$$\nabla f(\mathbf{x}) = 2\mathbf{A}\mathbf{x} + \mathbf{b}$$

Review: Gradient

$\nabla f(\mathbf{x})$ is the direction of fastest rate of increase of f at $\mathbf{x}$,

$$f(\mathbf{x} + \mathbf{d}) - f(\mathbf{x}) \approx \nabla f(\mathbf{x})^T \mathbf{d} \leq \|\nabla f(\mathbf{x})\| \cdot \|\mathbf{d}\|$$

where equality holds in the last step iff $\mathbf{d} = \alpha \nabla f(\mathbf{x})$ for some $\alpha \geq 0$.



Review: Chain rule

If $f : X \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $x_0 \in X$, $g : Y \subset \mathbb{R}^m \rightarrow \mathbb{R}^p$ is differentiable at $y_0 = f(x_0)$, then the composition of f and g defined by $h(x) = g(f(x))$ is differentiable at x_0 , and

$$h'(x_0) = g'(y_0)f'(x_0) = g'(f(x_0))f'(x_0)$$

Note. The order is important since $g'(y_0) \in \mathbb{R}^{p \times m}$ and $f'(x_0) \in \mathbb{R}^{m \times n}$ are matrices. In general $f'(x_0)g'(y_0)$ is **undefined**.

$$\begin{array}{ccccc} \mathbb{R}^n & \xrightarrow{f} & \mathbb{R}^m & \xrightarrow{g} & \mathbb{R}^p \\ x_0 & \mapsto & y_0 = f(x_0) & \mapsto & z_0 = h(x_0) = g(y_0) \\ \Delta x & \xrightarrow{f'} & \Delta y \approx f'(x_0)\Delta x & \xrightarrow{g'} & \Delta z \approx g'(y_0)\Delta y \approx g'(y_0)f'(x_0)\Delta x \end{array}$$

In component form,

$$[h'(x_0)]_{ij} = \frac{\partial h_i(x_0)}{\partial x_j} = \sum_{k=1}^m \frac{\partial g_i(y_0)}{\partial y_k} \cdot \frac{\partial f_k(x_0)}{\partial x_j} = \sum_{k=1}^m [g'(y_0)]_{ik} [f'(x_0)]_{kj}$$

Review: Chain rule

Example. $h(\mathbf{x}) = f(\mathbf{Ax} + \mathbf{b})$ has derivative $h'(\mathbf{x}_0) = f'(\mathbf{Ax}_0 + \mathbf{b})\mathbf{A}$. If f is real-valued,

$$\nabla h(\mathbf{x}_0) = \mathbf{A}^T [f'(\mathbf{Ax}_0 + \mathbf{b})]^T = \mathbf{A}^T \nabla f(\mathbf{Ax}_0 + \mathbf{b})$$

Example. Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{x}, \mathbf{d} \in \mathbb{R}^n$, define

$$g(t) = f(\mathbf{x} + t\mathbf{d})$$

Then

$$g'(t) = f'(\mathbf{x} + t\mathbf{d})\mathbf{d} = \nabla f(\mathbf{x} + t\mathbf{d})^T \mathbf{d} = \mathbf{d}^T \nabla f(\mathbf{x} + t\mathbf{d})$$

Note. g is the restriction of f to the straight line through $\mathbf{x}$ with direction $\mathbf{d}$. We can often get useful information about f by looking at g , which is usually easier to deal with.

First-order necessary condition

Consider unconstrained optimization problem, i.e. $X = \mathbb{R}^n$.

Theorem. If $\mathbf{x}^*$ is a local minimum of f and f is differentiable at $\mathbf{x}^*$, then its gradient at $\mathbf{x}^*$ vanishes, i.e.

$$\nabla f(\mathbf{x}^*) = \left[\frac{\partial f(\mathbf{x}^*)}{\partial x_1}, \dots, \frac{\partial f(\mathbf{x}^*)}{\partial x_n} \right]^T = \mathbf{0}.$$

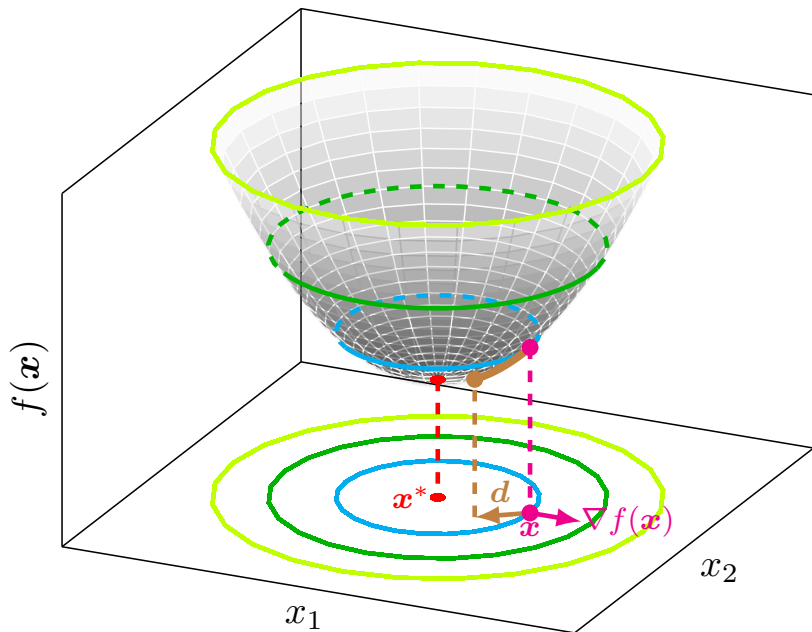
Proof. Let $\mathbf{d} \in \mathbb{R}^n$. Define $g(t) = f(\mathbf{x}^* + t\mathbf{d})$.

- Since $\mathbf{x}^*$ is a local minimum, $g(t) \geq g(0)$
- For $t > 0$,

$$\frac{g(t) - g(0)}{t} \geq 0 \implies g'(0) = \lim_{t \downarrow 0} \frac{g(t) - g(0)}{t} \geq 0$$

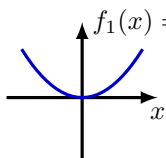
- By chain rule, $g'(0) = \sum_{i=1}^n d_i \frac{\partial f(\mathbf{x}^*)}{\partial x_i} = \mathbf{d}^T \nabla f(\mathbf{x}^*) \geq 0$
- Setting $\mathbf{d} = -\nabla f(\mathbf{x}^*) \implies \|\nabla f(\mathbf{x}^*)\|^2 \leq 0 \implies \nabla f(\mathbf{x}^*) = \mathbf{0}$

First-order Necessary Condition (cont'd)



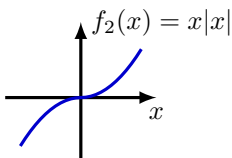
First-order Necessary Condition (cont'd)

A point $\mathbf{x}^*$ with $\nabla f(\mathbf{x}^*) = \mathbf{0}$ is called a **stationary point** of f .



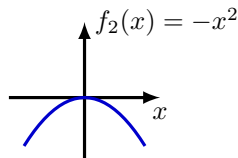
$$x^* = 0$$

minimum



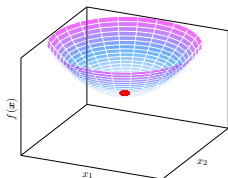
$$x^* = 0$$

inflection point



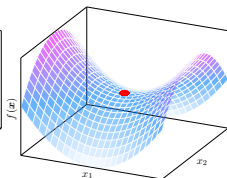
$$x^* = 0$$

maximum



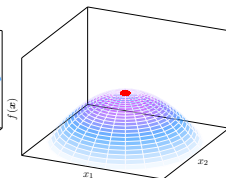
$$f(\mathbf{x}) = x_1^2 + x_2^2$$

minimum



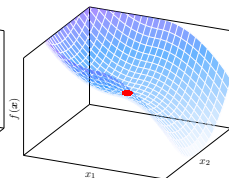
$$f(\mathbf{x}) = x_1^2 - x_2^2$$

saddle point



$$f(\mathbf{x}) = -x_1^2 - x_2^2$$

maximum



$$f(\mathbf{x}) = -x_1|x_1| + x_2^2$$

Note. Will see stationarity is sufficient for convex optimization.

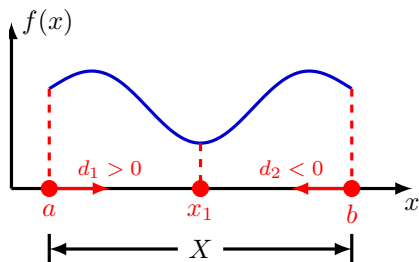
First-order Necessary Condition (cont'd)

For constrained optimization problem, i.e. $X \neq \mathbb{R}^n$,

- if $\mathbf{x}^*$ is in the **interior** of X , i.e. $B(\mathbf{x}^*, \epsilon) \subset X$ for some $\epsilon > 0$, then the proof still works, so $\nabla f(\mathbf{x}^*) = \mathbf{0}$
- otherwise, the proof shows $\mathbf{d}^T \nabla f(\mathbf{x}^*) \geq 0$ for any **feasible direction** $\mathbf{d}$ at $\mathbf{x}^*$
 - ▶ $\mathbf{d}$ is a feasible direction at $\mathbf{x} \in X$ if $\mathbf{x} + \alpha \mathbf{d} \in X$ for all sufficiently small $\alpha > 0$
- will revisit later

Example. $X = [a, b]$

- $f'(x_1) = 0$
- $d_1 f'(a) \geq 0 \implies f'(a) \geq 0$
- $d_2 f'(b) \geq 0 \implies f'(b) \leq 0$



Outline

- First-order conditions for unconstrained local min
- Second-order conditions for unconstrained local min

Review: Second derivative

The second-order partial derivatives of $f : X \subset \mathbb{R}^n \rightarrow \mathbb{R}$ at $\mathbf{x}_0 \in \text{int } X$ are

$$\frac{\partial^2 f(\mathbf{x}_0)}{\partial x_i \partial x_j}, \quad i, j = 1, 2, \dots, n$$

The **Hessian (matrix)** of f at $\mathbf{x}_0$, denoted by $\nabla^2 f(\mathbf{x}_0)$, is given by

$$[\nabla^2 f(\mathbf{x}_0)]_{ij} = \frac{\partial^2 f(\mathbf{x}_0)}{\partial x_i \partial x_j}, \quad i, j = 1, 2, \dots, n$$

Note. $\nabla^2 f$ is the Jacobian matrix of the vector-valued function ∇f .

If $\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}$ and $\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}$ exist in a neighborhood of $\mathbf{x}_0$ and are continuous at $\mathbf{x}_0$, then

$$\frac{\partial^2 f(\mathbf{x}_0)}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x}_0)}{\partial x_j \partial x_i}$$

so $\nabla^2 f(\mathbf{x}_0)$ is symmetric.

Will assume twice continuous differentiability when considering $\nabla^2 f$.

Review: Second derivative

Example. For an affine function $f(\mathbf{x}) = \mathbf{b}^T \mathbf{x} + c$

$$\nabla^2 f(\mathbf{x}) = \mathbf{0}$$

Example. For a quadratic function $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ with a symmetric $\mathbf{A}$,

$$\nabla^2 f(\mathbf{x}) = 2\mathbf{A}$$

Proof. Recall $\nabla f(\mathbf{x}) = 2\mathbf{A}\mathbf{x}$, so $\nabla^2 f(\mathbf{x}) = D(2\mathbf{A}\mathbf{x}) = 2\mathbf{A}$. In components,

$$\frac{\partial f(\mathbf{x})}{\partial x_j} = 2 \sum_{k=1}^n x_k A_{kj}$$

so

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = 2 \sum_{k=1}^n \frac{\partial x_k}{\partial x_i} A_{kj} = 2A_{ij}$$

Review: Chain rule for second derivative

The composition with affine function $g(\mathbf{x}) = f(\mathbf{A}\mathbf{x} + \mathbf{b})$ has Hessian

$$\nabla^2 g(\mathbf{x}) = \mathbf{A}^T \nabla^2 f(\mathbf{A}\mathbf{x} + \mathbf{b}) \mathbf{A}$$

Proof. Let $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$. Recall $\nabla g(\mathbf{x}) = \mathbf{A}^T \nabla f(\mathbf{y})$, so

$$\begin{aligned} \nabla^2 g(\mathbf{x}) &= D_{\mathbf{x}}(\nabla g(\mathbf{x})) = D_{\mathbf{x}}(\mathbf{A}^T \nabla f(\mathbf{y})) = \mathbf{A}^T D_{\mathbf{x}}(\nabla f(\mathbf{y})) \\ &= \mathbf{A}^T D_{\mathbf{y}}(\nabla f(\mathbf{y})) D_{\mathbf{x}} \mathbf{y} = \mathbf{A}^T \nabla^2 f(\mathbf{y}) \mathbf{A} \end{aligned}$$

In components,

$$\begin{aligned} \frac{\partial g(\mathbf{x})}{\partial x_j} &= \sum_k \frac{\partial f(\mathbf{y})}{\partial y_k} \frac{\partial y_k}{\partial x_j} = \sum_k \frac{\partial f(\mathbf{y})}{\partial y_k} A_{kj} \\ \frac{\partial^2 g(\mathbf{x})}{\partial x_i \partial x_j} &= \sum_k \frac{\partial}{\partial x_i} \frac{\partial f(\mathbf{y})}{\partial y_k} A_{kj} = \sum_k \sum_{\ell} \frac{\partial^2 f(\mathbf{y})}{\partial y_{\ell} \partial y_k} A_{\ell i} A_{kj} = [\mathbf{A}^T \nabla^2 f(\mathbf{y}) \mathbf{A}]_{ij} \end{aligned}$$

Special case. For $g(t) = f(\mathbf{x} + t\mathbf{d})$,

$$g''(t) = \mathbf{d}^T \nabla^2 f(\mathbf{x} + t\mathbf{d}) \mathbf{d}$$

Proof. Set $\mathbf{A} \leftarrow \mathbf{d}$, $\mathbf{x} \leftarrow t$, $\mathbf{b} \leftarrow \mathbf{x}$ in the general formula above.

Review: Second-order Taylor expansion

The second-order Taylor expansion for $g : \mathbb{R} \rightarrow \mathbb{R}$ takes the form

$$g(a + t) = g(a) + g'(a)t + \frac{1}{2}g''(a)t^2 + o(|t|^2) \quad (\text{T1})$$

The second-order Taylor expansion for $f : \mathbb{R}^n \rightarrow \mathbb{R}$ takes the form

$$f(\mathbf{x} + \mathbf{d}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} + o(\|\mathbf{d}\|^2) \quad (\text{T2})$$

i.e.

$$f(\mathbf{x} + \mathbf{d}) = f(\mathbf{x}) + \sum_{i=1}^n \frac{\partial f(\mathbf{x})}{\partial x_i} d_i + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} d_i d_j + o(\|\mathbf{d}\|^2)$$

Note. (T2) can be obtained by applying (T1) to $g(t) = f(\mathbf{x} + t\hat{\mathbf{d}})$ at $a = 0$ and $t = \|\mathbf{d}\|$, where $\hat{\mathbf{d}}$ is the unit vector in the direction $\mathbf{d}$, i.e. $\mathbf{d} = \|\mathbf{d}\|\hat{\mathbf{d}}$,

$$g(\|\mathbf{d}\|) = g(0) + g'(0)\|\mathbf{d}\| + \frac{1}{2}g''(0)\|\mathbf{d}\|^2 + o(\|\mathbf{d}\|^2)$$

By the chain rule, $g'(0) = \nabla f(\mathbf{x})^T \hat{\mathbf{d}}$, $g''(0) = \hat{\mathbf{d}}^T \nabla^2 f(\mathbf{x}) \hat{\mathbf{d}}$

Review: Second-order Taylor expansion

For a quadratic function $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$, the second-order Taylor expansion is exact with no $o(\|\mathbf{d}\|^2)$ term, i.e.

$$f(\mathbf{x} + \mathbf{d}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d}$$

Note. This can be used to find the expressions for ∇f and $\nabla^2 f$.

Assume $\mathbf{A}$ is symmetric; otherwise, replace $\mathbf{A}$ by $\tilde{\mathbf{A}} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$.

$$\begin{aligned} f(\mathbf{x} + \mathbf{d}) &= (\mathbf{x} + \mathbf{d})^T \mathbf{A} (\mathbf{x} + \mathbf{d}) + \mathbf{b}^T (\mathbf{x} + \mathbf{d}) + c \\ &= \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{d}^T \mathbf{A} \mathbf{x} + \mathbf{x}^T \mathbf{A} \mathbf{d} + \mathbf{d}^T \mathbf{A} \mathbf{d} + \mathbf{b}^T \mathbf{x} + \mathbf{b}^T \mathbf{d} + c \\ &= f(\mathbf{x}) + (2\mathbf{A} \mathbf{x} + \mathbf{b})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T (2\mathbf{A}) \mathbf{d} \end{aligned}$$

Comparison with the Taylor expansion shows that

$$\nabla f(\mathbf{x}) = 2\mathbf{A} \mathbf{x} + \mathbf{b}, \quad \nabla^2 f(\mathbf{x}) = 2\mathbf{A}.$$

Review: Postive definite matrices

A matrix $A \in \mathbb{R}^{n \times n}$ is **positive semidefinite**, denoted by $A \succeq O$, if

1. it is symmetric, i.e. $A = A^T$
2. $x^T A x \geq 0, \forall x \in \mathbb{R}^n$

It is **positive definite**, denoted by $A \succ O$, if condition 2 is replaced by

- 2'. $x^T A x > 0, \forall x \in \mathbb{R}^n$ and $x \neq 0$.

Note. For a quadratic form $x^T A x$, can always assume A is symmetric, since

$$x^T A x = x^T A^T x = x^T \left(\frac{A + A^T}{2} \right) x$$

A is **negative (semi)definite** if $-A$ is positive (semi)definite.

A is **indefinite** if it is neither positive semidefinite nor negative semidefinite, i.e. there exists $x_1, x_2 \in \mathbb{R}^n$ s.t.

$$x_1^T A x_1 > 0 > x_2^T A x_2$$

Review: Positive definite matrices

Example. $A = B^T B$ is positive semidefinite.

Proof. Obviously $A^T = A$. For any x ,

$$x^T A x = x^T B^T B x = (Bx)^T (Bx) = \|Bx\|^2 \geq 0$$

Example. $A = B^T B$ is positive definite iff B has full column rank.

Proof. Note

$$x^T A x = 0 \iff \|Bx\|^2 = 0 \iff Bx = \mathbf{0}$$

so

$$\begin{aligned} A \succ \mathbf{O} &\iff (x^T A x = 0 \iff x = \mathbf{0}) \\ &\iff (Bx = \mathbf{0} \iff x = \mathbf{0}) \\ &\iff B \text{ has full column rank} \end{aligned}$$

Review: Test for positive definiteness

A vector x is an **eigenvector** of a matrix A with associated **eigenvalue** λ if

$$Ax = \lambda x$$

We can find all eigenvalues by solving $\det(\lambda I - A) = 0$.

Theorem. Let A be a symmetric matrix.

- $A \succ O$ iff all its eigenvalues $\lambda > 0$.
- $A \succeq O$ iff all its eigenvalues $\lambda \geq 0$.

Exmaple. $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ is positive definite.

$$\det(\lambda I - A) = (\lambda - 1)(\lambda - 5) - 4 = 0 \implies \lambda = 3 \pm 2\sqrt{2} > 0$$

Exmaple. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is positive semidefinite.

$$\det(\lambda I - A) = (\lambda - 1)(\lambda - 4) - 4 = 0 \implies \lambda_1 = 0, \lambda_2 = 5$$

Review: Test for positive definiteness

Given matrix $A = (a_{ij}) \in \mathbb{R}^{n \times n}$, a $k \times k$ **principal submatrix** of A consists of k rows and k columns with the same indices $I = \{i_1 < i_2 < \cdots < i_k\}$,

$$A_I = \begin{bmatrix} a_{i_1 i_1} & \cdots & a_{i_1 i_k} \\ \vdots & \ddots & \vdots \\ a_{i_k i_1} & \cdots & a_{i_k i_k} \end{bmatrix}$$

A **principal minor of order k** of A is $\det A_I$ for some I with $|I| = k$.

If $I = \{1, 2, \dots, k\}$, $D_k(A) \triangleq \det A_I$ is called the **leading principal minor of order k** .

Theorem (Sylvester). Let A be a symmetric matrix.

- $A \succ \mathbf{0}$ iff $D_k(A) > 0$ for $k = 1, 2, \dots, n$.
- $A \succeq \mathbf{0}$ iff $\det A_I \geq 0$ for all $I \subset \{1, 2, \dots, n\}$

Note. For positive semidefiniteness, we need to check **all** principal minors, not just the leading principal minors.

Review: Test for positive definiteness

Exmaple. $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ is positive definite.

$$D_1(A) = \det(1) = 1 > 0, \quad D_2(A) = \det A = 1 > 0$$

Example. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is positive semidefinite.

$$D_1(A) = \det(1) = 1, \quad \det A_{\{2\}} = \det(4) = 4, \quad D_2(A) = \det A = 0$$

Note. It is **not** enough to check $D_k(A) \geq 0$ for all k !

Example. $A = \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}$ is negative semidefinite,

$$D_1(A) = \det(0) = 0, \quad D_2(A) = \det A = 0,$$

but

$$\det A_{\{2\}} = \det(-2) = -2 < 0$$

Review: Test for positive definiteness

Exmple. $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ is positive definite.

- Use definition,

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = x_1^2 + 4x_1x_2 + 5x_2^2 = (x_1 + 2x_2)^2 + x_2^2 \geq 0, \quad \forall \mathbf{x} \in \mathbb{R}^2$$

$$\text{with equality} \iff \begin{cases} x_1 + 2x_2 = 0 \\ x_2 = 0 \end{cases} \iff \mathbf{x} = \mathbf{0}$$

- Find eigenvalues by solving $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$

$$\det \begin{bmatrix} \lambda - 1 & -2 \\ -2 & \lambda - 5 \end{bmatrix} = (\lambda - 1)(\lambda - 5) - 4 = 0 \implies \lambda = 3 \pm 2\sqrt{2} > 0$$

- Check leading principal minors

$$D_1(\mathbf{A}) = \det(1) = 1 > 0, \quad D_2(\mathbf{A}) = \det \mathbf{A} = 1 > 0$$

Review: Test for positive definiteness

Exmample. $\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 8 \\ 1 & 8 & 1 \end{bmatrix}$ is not positive definite.

Check leading principal minors

$$D_1(\mathbf{A}) = \det(1) = 1 > 0, \quad D_2(\mathbf{A}) = \det \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = 1 > 0$$

$$D_3(\mathbf{A}) = \det \mathbf{A} = 1 \times \begin{vmatrix} 5 & 8 \\ 8 & 1 \end{vmatrix} - 2 \times \begin{vmatrix} 2 & 8 \\ 1 & 1 \end{vmatrix} + 1 \times \begin{vmatrix} 2 & 5 \\ 1 & 8 \end{vmatrix} = -36 < 0$$

Can also check eigenvalues, e.g. using `numpy.linalg.eig`,

$$\lambda_1 = 11.69585173, \quad \lambda_2 = 0.58307572, \quad \lambda_3 = -5.27892745$$

Review: Eigendecomposition

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ has the following eigendecomposition

$$A = Q\Lambda Q^T = \sum_{i=1}^n \lambda_i \mathbf{v}_i \mathbf{v}_i^T$$

where $\Lambda = \text{diag}\{\lambda_1, \dots, \lambda_n\}$, $Q = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ is an orthogonal matrix, i.e. $Q^T Q = Q Q^T = I$, and $A \mathbf{v}_i = \lambda_i \mathbf{v}_i$.

Example. $A = \frac{1}{4} \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$ has eigenvalues $\lambda_1 = \frac{1}{2}$ and $\lambda_2 = 1$, with corresponding eigenvectors $\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. The eigendecomposition is

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}^T + \begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}^T$$

Review: Eigendecomposition

The linear transformation $\mathbf{x} \mapsto \mathbf{y} = \mathbf{A}\mathbf{x} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T\mathbf{x}$ can be decomposed into three steps

$$\mathbf{x} \mapsto \tilde{\mathbf{x}} = \mathbf{Q}^T\mathbf{x} \mapsto \tilde{\mathbf{y}} = \mathbf{\Lambda}\tilde{\mathbf{x}} \mapsto \mathbf{y} = \mathbf{Q}\tilde{\mathbf{y}}$$

Recall $\mathbf{v}_1, \dots, \mathbf{v}_n$ form an orthonormal basis of $\mathbb{R}^n$,

1. Find the coordinate $\tilde{\mathbf{x}}$ of $\mathbf{x}$ in the basis $\mathbf{v}_1, \dots, \mathbf{v}_n$,

$$\mathbf{x} = \mathbf{Q}\tilde{\mathbf{x}} = \sum_{i=1}^n \tilde{x}_i \mathbf{v}_i, \quad \tilde{x}_i = \mathbf{v}_i^T \mathbf{x}$$

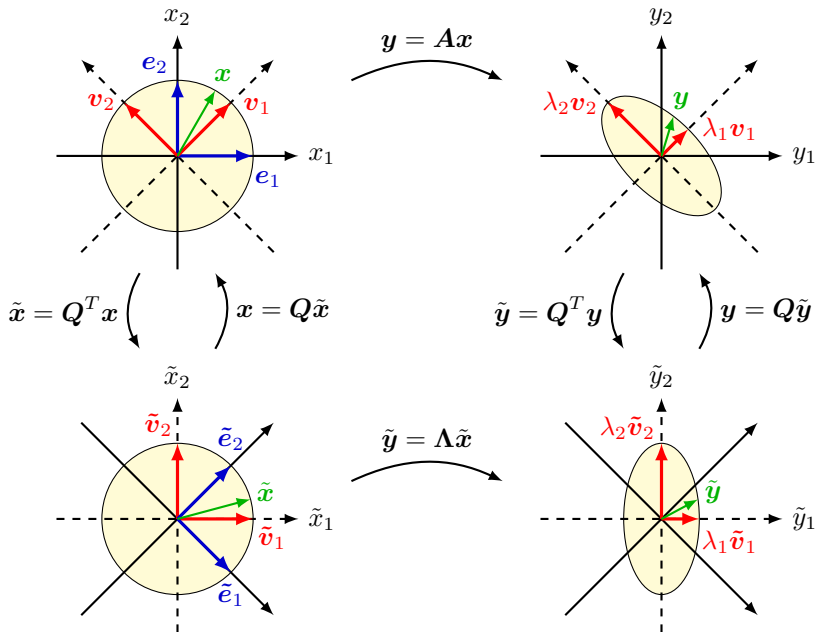
2. Scale the components of $\tilde{\mathbf{x}}$ by the corresponding eigenvalues,

$$\tilde{\mathbf{y}} = \mathbf{\Lambda}\tilde{\mathbf{x}}, \quad \tilde{y}_i = \lambda_i \tilde{x}_i$$

3. Find $\mathbf{y}$ from its coordinate $\tilde{\mathbf{y}}$ in the basis $\mathbf{v}_1, \dots, \mathbf{v}_n$,

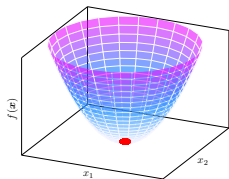
$$\mathbf{y} = \mathbf{Q}\tilde{\mathbf{y}} = \sum_{i=1}^n \tilde{y}_i \mathbf{v}_i$$

Review: Eigendecomposition



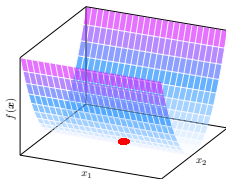
Review: Geometry of quadratic forms

Quadratic form $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ in $\mathbb{R}^2$



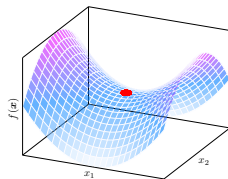
$$\mathbf{A} = \text{diag}\{1, 1\}$$

positive definite



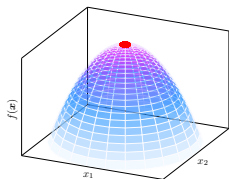
$$\mathbf{A} = \text{diag}\{0, 1\}$$

positive semidefinite



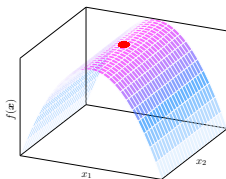
$$\mathbf{A} = \text{diag}\{1, -1\}$$

indefinite



$$\mathbf{A} = \text{diag}\{-1, -1\}$$

negative definite

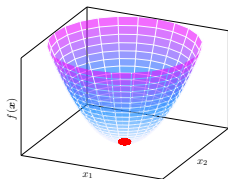


$$\mathbf{A} = \text{diag}\{-1, 0\}$$

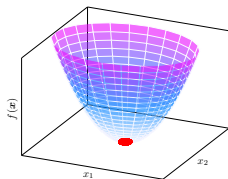
negative semidefinite

Review: Geometry of quadratic forms

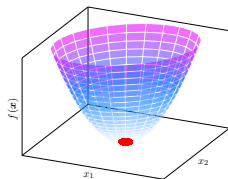
Quadratic form $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ in $\mathbb{R}^2$



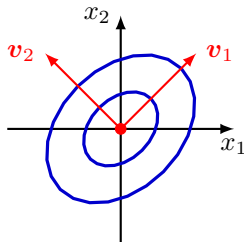
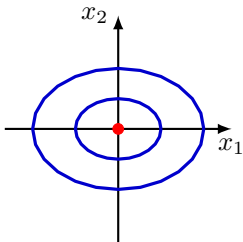
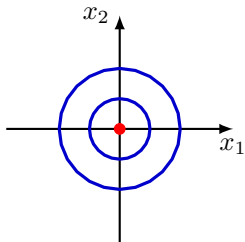
$$\mathbf{A} = \text{diag}\{1, 1\}$$



$$\mathbf{A} = \text{diag}\{\frac{1}{2}, 1\}$$



$$\mathbf{A} = \frac{1}{4} \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$



Review: Bounds on quadratic forms

Proposition. For a symmetric matrix $A \in \mathbb{R}^{n \times n}$,

$$\lambda_{\min} \|\mathbf{x}\|_2^2 \leq \mathbf{x}^T A \mathbf{x} \leq \lambda_{\max} \|\mathbf{x}\|_2^2, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest and the smallest eigenvalues of A , respectively.

Proof. Recall that A can be orthogonally diagonalized, i.e. $A = Q \Lambda Q^T$, where $\Lambda = \text{diag}\{\lambda_1, \dots, \lambda_n\}$ and $Q^T Q = I$. Let $\mathbf{x} = Q \tilde{\mathbf{x}}$.

$$\mathbf{x}^T A \mathbf{x} = \tilde{\mathbf{x}}^T (Q^T A Q) \tilde{\mathbf{x}} = \tilde{\mathbf{x}}^T \Lambda \tilde{\mathbf{x}} = \sum_{i=1}^n \lambda_i \tilde{x}_i^2 \leq \sum_{i=1}^n \lambda_{\max} \tilde{x}_i^2 = \lambda_{\max} \|\tilde{\mathbf{x}}\|_2^2$$

Then use the fact that orthogonal transformations preserve 2-norm, i.e.

$$\|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x} = (Q \tilde{\mathbf{x}})^T (Q \tilde{\mathbf{x}}) = \tilde{\mathbf{x}}^T (Q^T Q) \tilde{\mathbf{x}} = \tilde{\mathbf{x}}^T \tilde{\mathbf{x}} = \|\tilde{\mathbf{x}}\|_2^2.$$

Similarly for $\mathbf{x}^T A \mathbf{x} \geq \lambda_{\min} \|\mathbf{x}\|_2^2$.

Second-order necessary condition

Theorem. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable and $\mathbf{x}^*$ is a local minimum of f , then its **Hessian matrix** $\nabla^2 f(\mathbf{x}^*)$ is positive semidefinite, i.e.

$$\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0, \quad \forall \mathbf{d} \in \mathbb{R}^n$$

Proof. Fix $\mathbf{d} \in \mathbb{R}^n$. By the first-order necessary condition, $\nabla f(\mathbf{x}^*) = \mathbf{0}$. By the second-order Taylor expansion, for any $t > 0$,

$$f(\mathbf{x}^* + t\mathbf{d}) = f(\mathbf{x}^*) + \frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(t^2 \|\mathbf{d}\|^2) \geq f(\mathbf{x}^*)$$

So

$$\frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(\|\mathbf{d}\|^2) \geq 0, \quad \text{as } t \rightarrow 0$$

Taking the limit $t \rightarrow 0$ yields $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$.

Note. Can apply the same argument to $g(t) = f(\mathbf{x}^* + t\mathbf{d})$ with local minimum $t^* = 0$ and use chain rule to obtain $g''(0) = \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$.

Second-order sufficient condition

Theorem. Suppose f is twice continuously differentiable. If

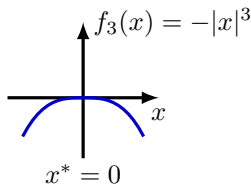
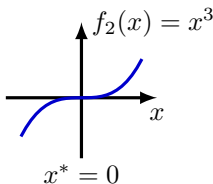
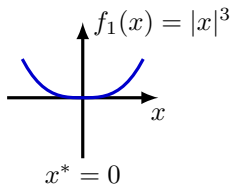
1. $\nabla f(\mathbf{x}^*) = 0$
2. $\nabla^2 f(\mathbf{x}^*)$ is positive definite, i.e.

$$\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} > 0, \quad \forall \mathbf{d} \neq \mathbf{0}$$

then $\mathbf{x}^*$ is a local minimum.

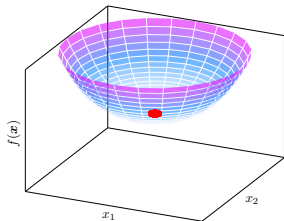
Proof. Use second-order Taylor expansion.

Note. In condition 2, positive definiteness **cannot** be replaced by positive semidefiniteness.

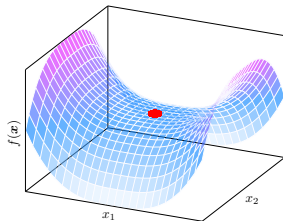


Second-order sufficient condition (cont'd)

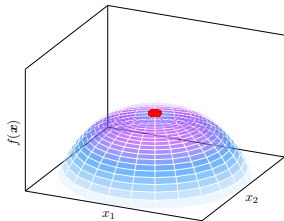
$\nabla f(\mathbf{0}) = \mathbf{0}$ and $\nabla^2 f(\mathbf{0}) = \mathbf{O}$ for all examples below.



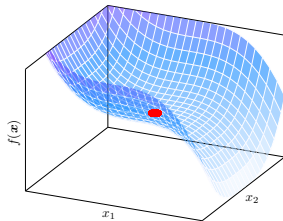
$$f(\mathbf{x}) = |x_1|^3 + |x_2|^3$$



$$f(\mathbf{x}) = |x_1|^3 - |x_2|^3$$



$$f(\mathbf{x}) = -|x_1|^3 - |x_2|^3$$



$$f(\mathbf{x}) = -x_1^3 + |x_2|^3$$